

Ömer Yasir Önal

Back-End & Applied AI Engineer

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SUMMARY

Recent Computer Engineering graduate and back-end & applied-AI engineer with hands-on experience building production-oriented systems, LLM integrations, and workflow automation. Comfortable across Node.js/Python backends, SQL, and multi-agent/agentive architectures, with React and native mobile (SwiftUI, Flutter) delivery. A fast learner who ships applicable solutions within real constraints. Looking for junior / entry-level full-time roles (and freelance) building AI agents, agentive workflows, and back-end systems.

SKILLS

Backend & APIs: Node.js, Express, TypeScript, FastAPI (Python), PostgreSQL, MySQL/Sequelize, SQLite, REST, JWT & bcrypt, WebSocket

AI & LLM Engineering: multi-agent orchestration, RAG (pgvector, ChromaDB), Model Context Protocol (MCP), Anthropic/OpenAI/Gemini/OpenRouter APIs, PyTorch (GPT/transformer from scratch), MLX LoRA fine-tuning, transfer learning (TensorFlow/Keras)

Frontend & Web: React 19, Next.js 16 (App Router), Astro 5, Tailwind CSS v4, Three.js / R3F, Vite

Mobile & Native: Swift 6 / SwiftUI, Flutter / Dart, Capacitor, Firebase, App Store delivery

DevOps & Infra: Docker, GitHub Actions CI/CD, Vercel, OCI self-hosting, pnpm monorepo, SwiftPM

Security & Trust: Ed25519 signing & build provenance, capability-token approval gates, OAuth 2.1 + Dynamic Client Registration, KVKK/consent compliance, honeypot & rate-limit hardening

EXPERIENCE

AI Fellow — Yapay Zeka ve Teknoloji Akademisi Istanbul · Dec 2025 – Present

- Selected among 1,500 of 31,700 applicants for the fellowship.
- Programme run by Google Türkiye, Girişimcilik Vakfı and T3 with the Ministry of Industry & Technology.
- Intensive curriculum in generative AI, ML/DL fundamentals, and end-to-end project development.

Software Engineer Intern — Exedra HRTech Istanbul · Jul 2025 – Aug 2025

- Built web product features and tests in a production HRTech environment.
- Shipped QRCheckIn as a side project.
- Worked in an Agile team with code review.

Instructor / Mentor — T3 Foundation — Deneyap Technology Workshops Istanbul · 2024 – 2025

- Taught and mentored students in the Deneyap technology workshops.

Researcher (lead contributor) — TÜBİTAK 2209-A Istanbul · Jun 2023 – Jun 2024

- Applied ML (XGBoost / Random Forest) to sensor data for structural / concrete durability.
- Work produced a paper accepted at IEEE SIU 2025.

Freelance Software Developer — Freelance Istanbul · May 2024 – Jan 2025

- Delivered end-to-end client web projects.
- Owned scope, timeline and SEO / Google Ads.

PROJECTS

AKIS — Verifiable AI software-development platform.

akisflow.com

TypeScript · Node.js · React 19 · PostgreSQL/pgvector · Docker · Multi-LLM · Ed25519 · MCP

UniSum — GPA & grade tracker, live on the App Store.

apps.apple.com/tr/app/unisum/id6742401580

Swift 6 · SwiftUI · Node.js · Express · MySQL · JWT & bcrypt · REST

newBrain — A Turkish LLM built from scratch.

github.com/OmerYasirOnal/newbrain

Python · PyTorch · MLX · FastAPI · Three.js · ChromaDB

A2 Reklam — Freelance corporate site, live in production.

a2reklam.com

Astro 5 · Tailwind CSS · PHP · Schema.org · GTM/GA4

UsageMeter — Private macOS menu-bar app for Claude usage.

omeryasironal.github.io/UsageMeter

Swift 6 · SwiftUI

Studio — Multi-agent game-development framework for Claude Code.

github.com/OmerYasirOnal/studio

TypeScript · Three.js · React Three Fiber · Capacitor · FastAPI

PUBLICATIONS

Machine learning for structural / concrete durability prediction (XGBoost / Random Forest) — IEEE SIU 2025, 2025

WRITING

Medium (@engomeryasironal) — What Are AI Agents? The Next Evolution in Coding · How Do AI Agents Work? (And Let's Build One) · AI Destekli Yazılım Geliştirme Rehberi: Hızlı ve Güvenli Adımlar

CERTIFICATIONS

- **Foundations of Project Management** — Google (Coursera), Jan 2026
- **Supervised Machine Learning: Regression and Classification** — DeepLearning.AI (Coursera), Nov 2025
- **Exploratory Data Analysis for Machine Learning** — Coursera, Aug 2023

EDUCATION

B.Sc. Computer Engineering — Fatih Sultan Mehmet Vakıf University — Graduated June 2026

LANGUAGES

Turkish (Native) · English (B2)